

(1.1) Dokažte, že v libovolném komplexním vektorovém prostoru V se skalárním součinem $\langle, \rangle$ platí (pro libovolné $\vec{u}, \vec{v} \in V$)

$$\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2),$$

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2),$$

kde $\operatorname{Re}(x)$ značí reálnou a $\operatorname{Im}(x)$ imaginární část komplexního čísla x .

ŘEŠENÍ:

$$\frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2) \stackrel{\text{z. definice normy vektoru}}{=} \frac{1}{4} (\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle) =$$

$$\stackrel{(SL4,2)}{\stackrel{(SSS)}}{=} \frac{1}{4} \left[\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle) \right] =$$

$$= \frac{1}{4} \left[2 \cdot (\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle) \right] \stackrel{(SSS)}{=} \frac{1}{2} (\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) =$$

$$= \frac{1}{2} (2 \cdot \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)) = \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

$$\downarrow \\ (a+bi) + (a-bi) = 2a \\ \text{neboli } x + \bar{x} = 2 \operatorname{Re}(x)$$

Jiným je 1. rovnost dokazána.

$$\frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) \stackrel{\text{definice normy}}{=} \frac{1}{4} (\langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle) =$$

$$\stackrel{(SL4,2)}{\stackrel{(SSS)}}{=} \frac{1}{4} \left[\langle \vec{u}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle - \langle \vec{u}, i\vec{v} \rangle - \langle i\vec{v}, \vec{u} \rangle - (\langle \vec{u}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle) \right] =$$

$$\begin{aligned}
 &= \frac{1}{4} \left[-2 \left(\langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle \right) \right] \stackrel{(S41)}{\stackrel{(SSS)}{=}} -\frac{1}{2} \left[i \langle \vec{u}, \vec{v} \rangle + \bar{i} \langle \vec{v}, \vec{u} \rangle \right] = \\
 &\stackrel{(SSS)}{=} -\frac{1}{2} i \left(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} \right) = \left(-\frac{1}{2} i \right) \cdot (2i) \cdot \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \\
 &\quad \downarrow \\
 &\quad (a+bi) - (a-bi) = 2bi \\
 &\quad \text{neboli } x - \bar{x} = 2i \cdot \operatorname{Im}(x)
 \end{aligned}$$

Tím je dokázána i druhá rovnost.